Technical Specification - Rigidity Inference from Trajectories (RIFT)

Version 1.0

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1 Overview

Rigidity Inference from Trajectories (RIFT) is a trajectory-based statistical framework for detecting structural imposture in cellular broadcast systems.

RIFT operates on receiver-observable evidence rather than unobservable ground truth. In particular, RIFT models how broadcast-related observations evolve over time under environmental coupling and receiver constraints, then detects deviations from empirically learned legitimate behavior.

RIFT proceeds as follows:

- Collect receiver observations over time and form trajectory segments.
- Align observations to a fixed sampling interval.
- Convert each segment into a discrete identity symbol stream and RF measurement stream.
- Extract trajectory-structured features (entropy, transition structure, complexity, and RF coupling).
- Condition features on RF regime to remove confounding.
- Model the manifold of legitimate residual behavior from a trusted baseline.
- Score and flag segments that are both statistical outliers and directionally consistent with rigidity under IRP.

RIFT supports:

- Offline capture analysis (e.g., PCAP, QMDL files).
- Live streaming telemetry using sliding windows.

2 Scope

This document specifies the Rigidity Inference from Trajectories protocol (RIFT), Version 1.0.

RIFT is a trajectory-based statistical method for detecting structural imposture in cellular broadcast systems by:

- Observing broadcast-related information over time.
- Extracting trajectory-structured features from those observations.

- Conditioning those features on radio-frequency (RF) environment to remove confounding.
- Modeling a baseline manifold of legitimate behavior.
- Detecting statistically significant deviations consistent with impersonation constraints.

RIFT supports offline capture analysis (e.g., PCAP, QMDL) and live streaming telemetry using sliding-window evaluation.

RIFT specifies a detection protocol. It does not specify RF capture hardware, baseband access methods, or cellular protocol decoding implementations.

3 Normative Language

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, and “MAY” in this document are to be interpreted as described in RFC 2119.

4 Definitions and Notation

This section defines common terms and symbols used throughout this specification.

4.1 Definitions

- **Base station (cell):** A cellular network node broadcasting system information and serving user equipment (UE).
- **Broadcast identity state:** The set of identity-relevant broadcast parameters emitted by the base station (e.g., PLMN, TAC, Cell ID, PCI, MIB/SIB).
- **Receiver observation:** The set of broadcast-related elements successfully decoded by the receiver at a given time, including RF measurements.
- **Trajectory segment:** A time-ordered sequence of receiver observations over an interval, used as the evidentiary object for analysis.
- **Baseline:** A trusted set of trajectory segments believed to represent legitimate network behavior, used to fit conditioning models and the legitimacy manifold.

4.2 Notation

Symbol	Meaning
t	Time index (seconds).
Δt	Fixed sampling interval used for time alignment (seconds).
$B(t)$	True broadcast identity state of a base station at time t.
$Y(t)$	Receiver observation at time t.
$E(t)$	Environmental RF state at time t (fading, interference, mobility).
$R_{\text{sys}}(t)$	Receiver internal state at time t (hardware, firmware, orientation, decode success).
$\mathcal{T}_{a:b}$	Trajectory segment consisting of observations $\{Y(t)\}$ from time a to b.
S_t	Canonical identity symbol derived from $Y(t)$ at aligned time step t.
SINR(t)	Signal-to-Interference-plus-Noise Ratio at time t (dB).
RSRP(t)	Reference Signal Received Power at time t (dBm).
Z_i	Conditioning vector for trajectory segment i.
F_i	Feature vector extracted from trajectory segment i.
$m_j(\cdot)$	Continuous conditioning function for feature dimension j.
R_i	Residual feature vector for trajectory segment i after conditioning.
μ	Mean residual vector computed from baseline residuals.
Σ	Residual covariance matrix computed from baseline residuals.
D_i	Mahalanobis distance of residual R_i from baseline manifold.
τ	Baseline-derived threshold for D_i (e.g., 99th percentile).
w	Direction vector encoding rigidity-consistent feature signs.
ρ_i	Directional rigidity score $w^T R_i$.
γ	Baseline-derived threshold for ρ_i (e.g., 5th percentile).

All symbols are re-defined in context in subsequent sections to preserve readability for implementors.

5 System Model

This section defines the relationship between (a) the base station's true broadcast identity state and (b) what a receiver can observe. RIFT is defined over receiver-observable evidence, not over unobservable ground truth.

5.1 True Broadcast Identity State

Let t denote time in seconds.

Let $B(t)$ denote the **true broadcast identity state** of a base station at time t.

In plain terms, $B(t)$ represents what the base station is actually broadcasting at that time, including identity-related broadcast parameters such as:

- Public Land Mobile Network (PLMN) identifiers.
- Tracking Area Code (TAC).
- Cell Identifier (Cell ID).
- Frequency and band.
- Physical Cell Identifier (PCI).
- Broadcast system information blocks (e.g., MIB/SIB content).
- Neighbor cell references and reselection parameters.

RIFT makes no assumption that $B(t)$ changes frequently. In many real deployments, $B(t)$ is piecewise constant and changes only when the network is reconfigured.

5.2 Receiver Observation

A receiver cannot directly observe $B(t)$. Instead, at time t it records an **observation** denoted $Y(t)$.

$Y(t)$ is the set of broadcast-related information successfully decoded at time t, and typically includes:

- Serving cell identity parameters (e.g., frequency, PCI, TAC, Cell ID) as decoded.
- Decoded broadcast blocks (e.g., MIB/SIB elements) available at that time.
- RF measurements (at minimum SINR; optionally RSRP and others).

RIFT models the observation process as:

$$Y(t) = h\big(B(t), E(t), R_{\text{sys}}(t)\big)$$

where:

- $h(\cdot)$ is the receiver's observation mechanism (decoding, measurement, and reporting).

- $E(t)$ is the environmental RF state at time t (fading, interference, mobility, multipath).
- $R_{sys}(t)$ is the receiver internal state at time t (hardware, firmware, antenna orientation, decode success).

This equation is not used to compute $B(t)$. It states that even if $B(t)$ is constant, $Y(t)$ can vary due to environment and receiver state.

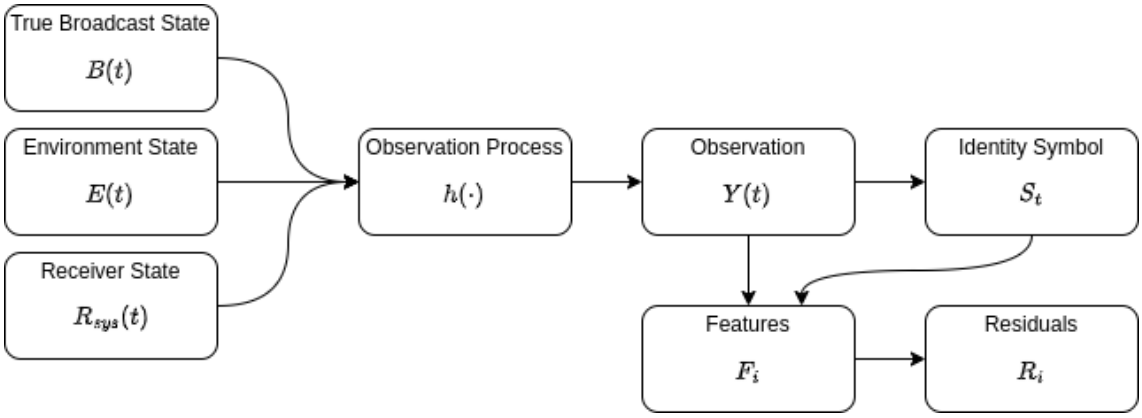

Figure 5.2-a. RIFT System Observation Model and Evidence Flow (*$B(t)$*, *$E(t)$*, *$R_{sys}(t)$*, and *$Y(t)$*).

5.3 Trajectory Segment

RIFT operates on **trajectory segments**, which are time-ordered sequences of observations.

Let a be a start time and b be an end time (seconds). Define the trajectory segment:

$$\mathcal{T}_{a:b} = \{Y(t)\}_{t=a}^{b}$$

where:

- $\mathcal{T}_{a:b}$ is the evidentiary object used for feature extraction and scoring.
- $Y(t)$ is the receiver observation at time t.

Under Stateful Temporal Entropy (STE), legitimacy claims MUST be supported by trajectory evidence over irreversible progression. RIFT treats $\mathcal{T}_{a:b}$ as that evidence object.

6 Time Alignment

Many data sources (live telemetry, baseband logs, PCAP-derived events) are irregularly sampled. However, entropy and transition measures require a consistent time base. This section specifies how observations MUST be aligned to a uniform sampling interval.

6.1 Sampling Interval

Let Δt denote the fixed sampling interval in seconds.

Implementations MUST choose $\Delta t > 0$ and align observations onto times $t = k\Delta t$ for integer k.

Recommended:

$$\Delta t = 1 \text{ second}$$

6.2 Alignment Procedure

For each aligned time step $t = k\Delta t$:

- If multiple raw observations occur within $(t - \Delta t, t]$, RF measurements (e.g., SINR, RSRP) MUST be aggregated using the **median**.
- If multiple identity observations occur within $(t - \Delta t, t]$, the identity symbol (defined in Section 7) MUST use the **most recently decoded** value in that interval.
- If no observation is available for $(t - \Delta t, t]$, the aligned sample MUST be marked as missing.

This procedure performs **time alignment** only. It MUST NOT be interpreted as statistical binning of RF values or features.

6.3 Missingness Requirements

Let T be the number of aligned time steps in a trajectory segment.

Minimum Segment Length. Implementations MUST enforce a minimum aligned segment length $T_{\min}$. Entropy, transition, and mutual information estimators are unstable for very small T. A recommended value is:

$$T_{\min} \geq 300$$

Segments with $T < T_{\min}$ MUST be discarded or labeled invalid.
Implementations MUST define a missingness policy. A common policy is:

- A segment is valid only if at least $p_{\min}$ of aligned samples are present (not missing).

Where:

- $p_{\min}$ is a configured minimum fraction in $[0, 1]$.

Recommended:

$$p_{\min} = 0.7$$

Segments failing missingness requirements MUST be discarded or flagged as invalid for scoring.

7 Identity Symbol Construction

RIFT represents broadcast identity observations as a discrete symbol stream. This section specifies how the symbol stream MUST be constructed from aligned receiver observations.

7.1 Canonical Identity Symbol

Let t denote an aligned time step after applying Section 6.
Let $Y(t)$ denote the receiver observation at time t.
Implementations MUST derive a canonical identity symbol S_t from $Y(t)$.
The preferred canonical symbol is:

$$S_t = \left(\text{frequency}_t, \text{PCI}_t, \text{TAC}_t, \text{cellID}_t\right)$$

where:

- frequency_t is the downlink frequency (or equivalent channel identifier) decoded at time t.
- PCI_t is the Physical Cell Identifier decoded at time t.
- TAC_t is the Tracking Area Code decoded at time t.
- cellID_t is the Cell Identifier decoded at time t.

If one or more fields are unavailable, implementations MAY use a reduced symbol such as $(\text{frequency}_t, \text{PCI}_t)$, but MUST document the chosen symbol definition and apply it consistently.

7.2 Identity Stream

For a trajectory segment containing T aligned time steps, the identity stream is:

$$S_1, S_2, \ldots, S_T$$

This discrete sequence is the primary input to the entropy, transition, and complexity measures defined in Section 9.

7.3 Interpretation

RIFT does not assume the true broadcast identity $B(t)$ changes frequently. The symbol stream S_t reflects what was observed and decoded under environmental coupling and receiver limitations. Structural rigidity is inferred from the trajectory properties of S_t (and its relationship to RF measurements), not from any assumption of frequent ground-truth identity changes.

8 RF Metrics and Conditioning

RF conditions influence what is observed and decoded. Without conditioning, strong-signal stability may be mistaken for rigidity, and weak-signal instability may be mistaken for diversity. This section defines RF metrics and the conditioning vector used to remove RF-driven confounding.

8.1 Signal-to-Interference-plus-Noise Ratio (SINR)

Let t denote an aligned time step.

Let $P_{\text{signal}}(t)$ denote received signal power at time t.

Let $P_{\text{interference}}(t)$ denote received interference power at time t.

Let $P_{\text{noise}}(t)$ denote receiver noise power at time t.

Define SINR as:

$$\text{SINR}(t) = \frac{P_{\text{signal}}(t)}{P_{\text{interference}}(t) + P_{\text{noise}}(t)}$$

SINR is typically expressed in decibels (dB). Implementations MUST treat SINR as a scalar RF quality metric.

Interpretation:

- High SINR implies a relatively clean channel.
- Low SINR implies a noisy or contested channel and tends to increase decoding variability.

8.2 Reference Signal Received Power (RSRP)

Let t denote an aligned time step.

Define RSRP(t) as the Reference Signal Received Power (typically expressed in dBm), representing absolute received signal strength for reference signals.

RSRP helps distinguish:

- Strong-but-interfered conditions (high RSRP, low SINR).
- Weak-but-clean conditions (low RSRP, high SINR).

8.3 Conditioning Vector

RIFT conditions features on RF regime using summary statistics computed per trajectory segment.

Let i denote a trajectory segment.

Let $\{\mathrm{SINR}(t)\}$ be the SINR samples observed over that segment.

Let $\{\mathrm{RSRP}(t)\}$ be the RSRP samples observed over that segment.

Define:

- $\mathrm{SINR}_{\mathrm{median}}$ as the median SINR across the segment.
- Q_{75}^{SINR} as the 75th percentile of SINR across the segment.
- Q_{25}^{SINR} as the 25th percentile of SINR across the segment.
- $\mathrm{SINR}_{\mathrm{IQR}} = Q_{75}^{\mathrm{SINR}} - Q_{25}^{\mathrm{SINR}}$ as the SINR interquartile range (IQR).
- $\mathrm{RSRP}_{\mathrm{median}}$ as the median RSRP across the segment.

The conditioning vector for segment i is:

$$Z_i = \left(\mathrm{SINR}_{\mathrm{median}}, \mathrm{SINR}_{\mathrm{IQR}}, \mathrm{RSRP}_{\mathrm{median}}\right)$$

Purpose:

- To compare segments under similar RF quality and variability.
- To remove expected, non-adversarial effects of RF on identity trajectory structure.

Implementations MAY include additional RF quality metrics (e.g., RSRQ) in the conditioning vector provided multicollinearity is addressed and conditioning stability is preserved.

9 Feature Extraction

RIFT converts each trajectory segment into a feature vector capturing structural properties of the observed identity stream and its coupling to RF changes.

Let i denote a trajectory segment.

Let the segment contain T aligned time steps.

Let S_t denote the identity symbol at aligned time step t (Section 7).

Let SINR(t) denote the SINR measurement at aligned time step t (Section 8.1).

The feature vector for segment i is:

$$F_i = \left[H_S,\ H_{\text{rate}},\ H_{\text{LZ}},\ T_{\text{sparse}},\ r,\ I_{\text{coupling}}\right]$$

where each component is defined below.

Logarithm Base. All logarithms in this specification (entropy, entropy rate, and mutual information) use the natural logarithm $\log(\cdot) = \ln(\cdot)$ unless otherwise stated. Units are therefore in nats. Implementations MAY use base-2 logarithms provided this choice is applied consistently across baseline training and evaluation.

9.1 Identity Entropy H_S

What it measures. Identity entropy measures the diversity of observed identity symbols within the segment.

Zero-Probability Convention. By convention, terms with $p(s) = 0$ are treated as contributing zero to the entropy sum (i.e., $0 \log 0 = 0$). Implementations MUST avoid numerical evaluation of log(0) by excluding zero-probability terms from the summation.

Definitions.

- Let $\mathcal{S}$ be the set of unique identity symbols observed in the segment.
- Let $p(s)$ be the empirical probability of symbol $s \in \mathcal{S}$, defined as the fraction of time steps where $S_t = s$.

Formally:

$$p(s) = \frac{\#\{t \in \{1, \dots, T\} : S_t = s\}}{T}$$

Equation. Shannon entropy is:

$$H_S = -\sum_{s \in \mathcal{S}} p(s)\ \log p(s)$$

where $\log(\cdot)$ is the natural logarithm.

Interpretation.

- Higher H_S indicates more diverse observed identity states.
- Lower H_S indicates constrained identity observations.

9.2 Entropy Rate H_{rate} (First-Order Markov)

What it measures. Entropy rate measures unpredictability of identity transitions and distinguishes rich transition structure from simple repetition or oscillation.

Definitions.

- Let $\mathcal{S}$ be the set of unique symbols.
- Let $p(s_i)$ be the empirical probability of symbol s_i as defined above.
- Let $P(s_j \mid s_i)$ be the empirical transition probability from s_i to s_j.

Empirical transition probability:

$$P(s_j \mid s_i) = \frac{\#\{t \in \{1, \dots, T-1\} : S_t = s_i,\ S_{t+1} = s_j\}}{\#\{t \in \{1, \dots, T-1\} : S_t = s_i\}}$$

Equation.

$$H_{\text{rate}} = -\sum_{s_i \in \mathcal{S}} \sum_{s_j \in \mathcal{S}} p(s_i)\, P(s_j \mid s_i)\, \log P(s_j \mid s_i)$$

Interpretation.

- Higher H_{rate} indicates richer, less predictable transition structure.
- Lower H_{rate} indicates constrained transitions or repetitive patterns.

9.3 Lempel–Ziv Complexity H_{LZ}

What it measures. Lempel–Ziv (LZ) complexity estimates algorithmic compressibility of the identity sequence without assuming a Markov model.

Definitions.

- Let $LZ(S)$ be the Lempel–Ziv parsing complexity of the sequence $S_1, \dots, S_T$, defined as the number of distinct phrases produced by standard LZ parsing.

- Let T be the sequence length.

Equation. A common normalization is:

$$H_{\mathrm{LZ}} = \frac{LZ(S)}{T/\log T}$$

Interpretation.

- Lower H_{LZ} indicates a more compressible (more structured) sequence.
- Higher H_{LZ} indicates less compressible (more irregular) behavior.

9.4 Transition Sparsity $T_{\mathbf{sparse}}$

What it measures. Transition sparsity measures how densely connected the observed transition graph is.

Definitions.

- Let V be the number of unique symbols observed, i.e., $V = |\mathcal{S}|$.
- Let E be the number of unique directed transitions ($s_i \to s_j$) observed at least once.

The maximum number of possible directed transitions is V^2 (including self-transitions).

Equation.

$$T_{\mathrm{sparse}} = \frac{E}{V^2}$$

Interpretation.

- Lower T_{sparse} indicates constrained transition structure.
- Higher T_{sparse} indicates richer connectivity among observed states.

9.5 Change Rate r

What it measures. Change rate measures how frequently identity changes occur across aligned time steps.

Definitions.

- Define the indicator $I_t = 1$ if $S_t \neq S_{t-1}$, else $I_t = 0$.
- T is the number of time steps.

Equation.

$$r = \frac{1}{T-1} \sum_{t=2}^{T} I_t$$

Interpretation.

- Higher r indicates more frequent identity changes.
- Lower r indicates more stable identity over time.

Transition-based feature components characterize the structural richness of the identity sequence. Legitimate broadcast environments typically induce multi-state transition graphs with diverse connectivity. Under impersonation constraints (IRP), admissible transition structure becomes reduced and highly constrained. Figure 9.5-a illustrates this contrast conceptually.

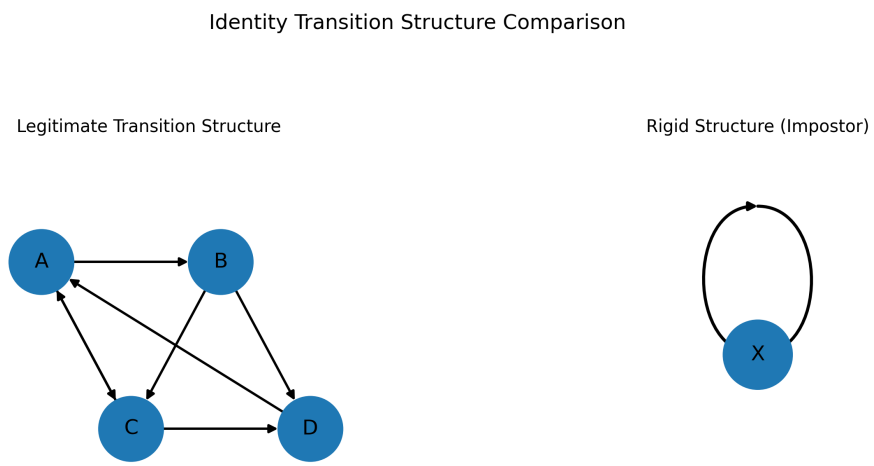

Figure 9.5-a. Identity transition structure Concept comparison illustrating structural richness versus rigidity.

9.6 Information-Theoretic Coupling I_{coupling}

What it measures. Coupling measures statistical dependence between RF changes and identity transitions. This captures exogenous coupling emphasized by STE. Terms with $p(x, y) = 0$ are omitted from the summation.

Definitions.

- Let SINR(t) be the SINR at time step t.
- Define $\Delta\text{SINR}_t = \left|\text{SINR}(t) - \text{SINR}(t-1)\right|$ for $t \geq 2$.
- Define $\Delta S_t = I_t$ for $t \geq 2$, where I_t is the identity-change indicator above.

RIFT defines coupling using mutual information between the random variables ΔSINR and ΔS derived from the sequences $\{\Delta\text{SINR}_t\}$ and $\{\Delta S_t\}$.

Mutual information definition. Let X and Y be random variables with joint distribution $p(x, y)$ and marginals $p(x)$ and $p(y)$. Mutual information is:

$$I(X;Y) = \sum_x \sum_y p(x,y) \log\left(\frac{p(x,y)}{p(x)\,p(y)}\right)$$

RIFT coupling score.

$$I_{\text{coupling}} = I(\Delta\text{SINR}; \Delta S)$$

Estimation requirement. Implementations MUST specify a mutual-information estimator. Because ΔS is binary and ΔSINR is continuous, a compliant approach is:

- Discretize ΔSINR into K quantile-based bins computed from the empirical distribution of ΔSINR within the segment under evaluation, then compute empirical $p(x, y)$.

Where:

- K is a configured integer (recommended $K = 8$).

Interpretation.

- Higher I_{coupling} indicates identity transitions are dependent on RF changes.
- Lower I_{coupling} indicates identity transitions are less coupled to RF changes, consistent with rigidity under IRP.

Discretization for mutual information estimation applies only to ΔSINR for the purpose of MI computation and does not constitute feature binning in the conditioning framework.

10 Continuous Conditioning

Features such as entropy and transition rate naturally vary with RF regime. RIFT removes these expected effects by conditioning each feature on the RF-conditioning vector Z_i computed per segment.

10.1 Conditioning Model

Let i denote a trajectory segment.

Let $Z_i = \left(\mathrm{SINR}_{\mathrm{median}}, \mathrm{SINR}_{\mathrm{IQR}}, \mathrm{RSRP}_{\mathrm{median}}\right)$ be the conditioning vector for segment i (Section 8.3).

Let $F_i = [F_{i1}, \dots, F_{id}]$ denote the d-dimensional feature vector for segment i (Section 9), where in RIFT v1.0:

- $d = 6$
- $F_{i1} = H_S,\ F_{i2} = H_{\mathrm{rate}},\ F_{i3} = H_{\mathrm{LZ}},\ F_{i4} = T_{\mathrm{sparse}},\ F_{i5} = r,\ F_{i6} = I_{\mathrm{coupling}}$

For each feature dimension $j \in \{1, \dots, d\}$, RIFT fits a smooth regression function:

$$F_{ij} = m_j(Z_i) + \epsilon_{ij}$$

where:

- $m_j(\cdot)$ is a continuous conditioning function (e.g., a spline-based Generalized Additive Model).
- ϵ_{ij} is the residual variation not explained by RF regime.

The regression function $m_j(\cdot)$ captures the expected value of feature component F_{ij} under RF conditioning variable Z_i. The residual ϵ_{ij} represents the deviation of the observed feature from this expected value. Only the residual component is retained for legitimacy manifold modeling, ensuring that RF-driven structural effects are removed prior to anomaly detection. Figure 10.1-a illustrates this conditioning process conceptually.

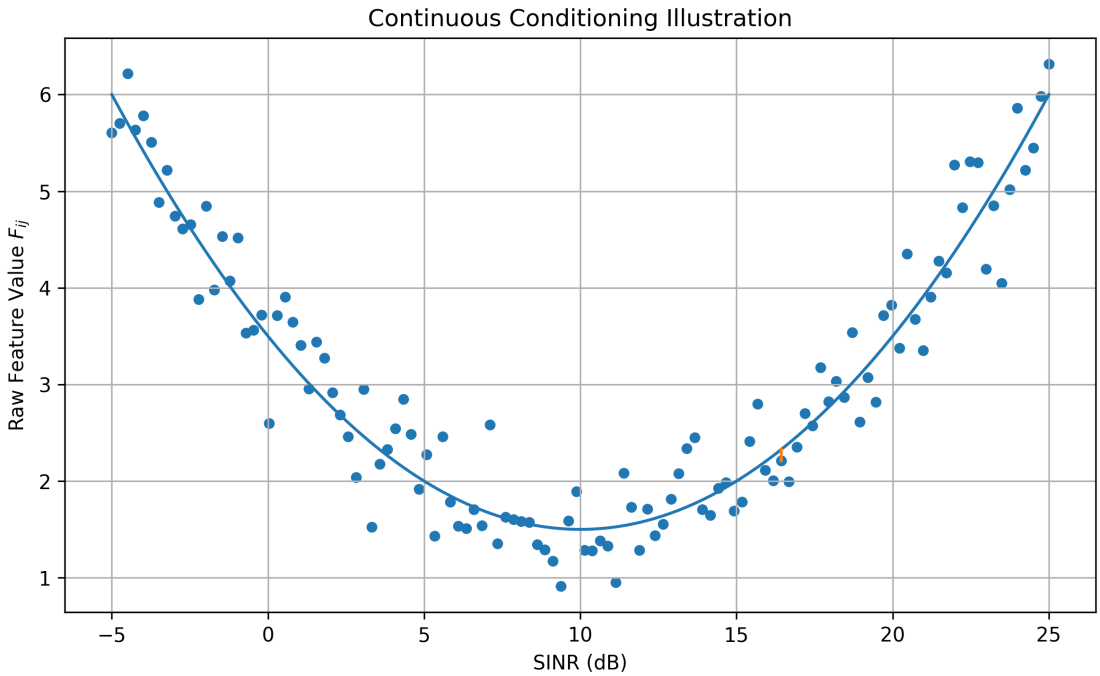

Figure 10.1-a. Continuous conditioning Concept of feature component F_{ij} on RF state Z_i, with residual ϵ_{ij}.

10.2 Residual Feature Vector

Let $\widehat{m}_j(Z_i)$ denote the fitted model prediction for feature j given Z_i.

Define the residual for dimension j as:

$$R_{ij} = F_{ij} - \widehat{m}_j(Z_i)$$

Define the residual feature vector:

$$R_i = [R_{i1}, R_{i2}, \dots, R_{id}]$$

10.3 Requirements

- Conditioning models $m_j(\cdot)$ MUST be fit using baseline segments only (Section 8).
- Once fit, conditioning models MUST be treated as fixed during evaluation on non-baseline data.
- Implementations MUST store the fitted model parameters sufficient to reproduce $\widehat{m}_j(Z)$ for any segment.

10.4 Rationale

Conditioning prevents the detector from confusing:

- Low entropy due to high SINR (legitimate stability) with rigidity.
- High entropy due to low SINR (legitimate instability) with diversity.

By operating on R_i, RIFT measures structural deviation beyond what RF regime alone predicts.

11 Legitimacy Manifold Modeling

RIFT models legitimate behavior as an empirical manifold in residual feature space. This manifold is learned from baseline segments believed to represent legitimate networks.

11.1 Baseline Dataset

Let the baseline dataset consist of n trajectory segments, indexed by $i \in \{1, \dots, n\}$.

Minimum Baseline Size. Let n denote the number of baseline segments. Reliable covariance estimation requires $n \gg d$.

Implementations SHOULD enforce:

$$n \geq 10d$$

where d is the feature dimension.

For each baseline segment i:

- Compute feature vector F_i (Section 9).
- Compute conditioning vector Z_i (Section 8.3).
- Compute residual vector R_i using fitted conditioning models (Section 10).

Baseline segments MUST be collected from trusted environments. If baseline integrity is uncertain, RIFT results are not meaningful.

Figure 11.1-a shows a three-dimensional projection for visualization; RIFT operates in the full d-dimensional residual feature space.

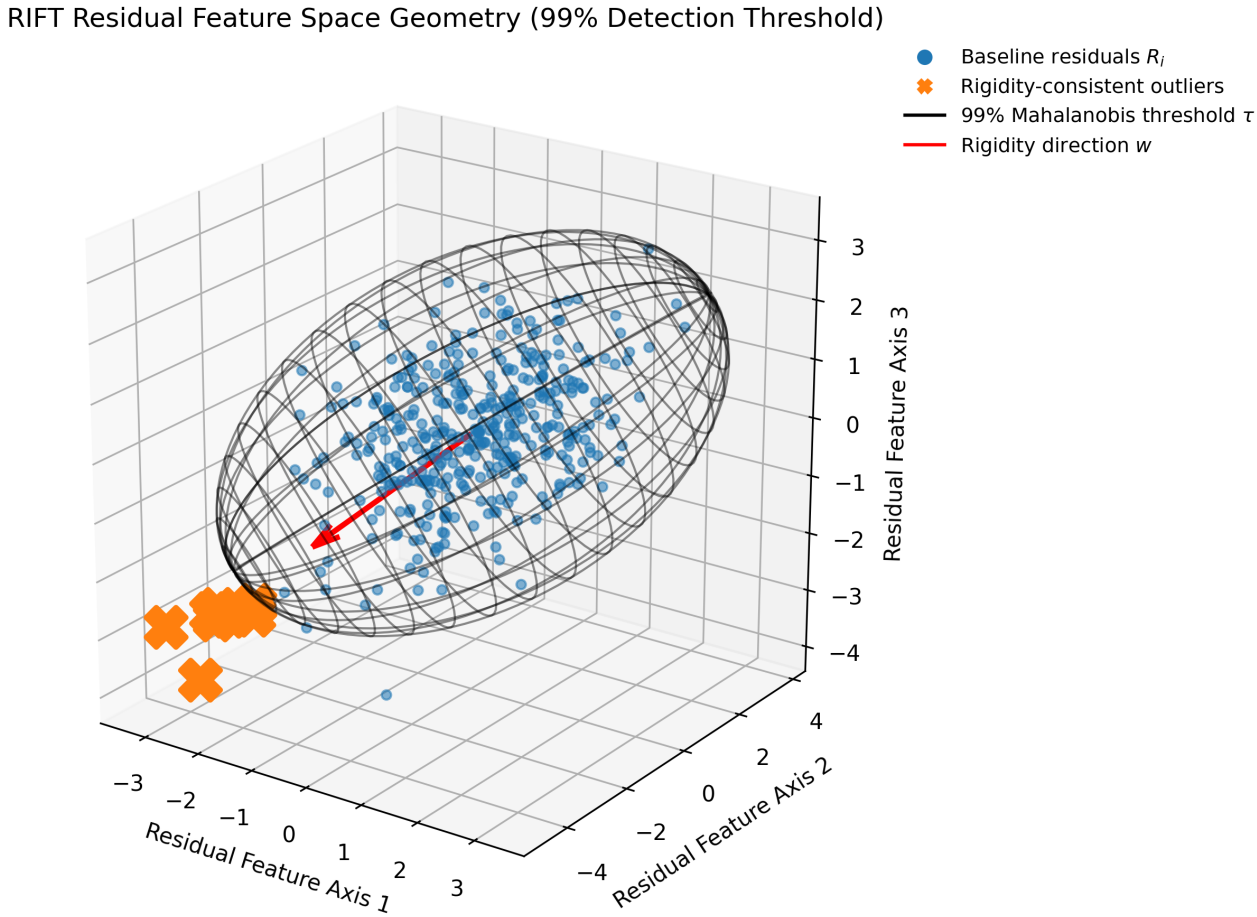

Figure 11.1-a. Residual Feature Space Geometry Concept at the 99% Mahalanobis Threshold (τ).

11.2 Mean and Covariance

Let $R_i \in R^d$ denote the d-dimensional residual vector for baseline segment i.

Define the baseline mean residual vector:

$$\mu = \frac{1}{n}\sum_{i=1}^{n} R_i$$

where:

- $\mu \in R^d$.

Define the baseline covariance matrix:

$$\Sigma = \frac{1}{n-1}\sum_{i=1}^{n}(R_i - \mu)(R_i - \mu)^T$$

where:

- $\Sigma \in R^{d \times d}$.
- $(\cdot)^T$ denotes vector transpose.

The empirical pair (μ, Σ) defines the geometric structure of the baseline residual distribution. Figure 11.1-a illustrates this geometry conceptually in three dimensions.

11.3 Robust Estimation Requirements

Covariance inversion (Section 12.1) requires Σ to be non-singular.

Implementations MUST address potential numerical issues:

- If Σ is singular or ill-conditioned, implementations MUST use one or more of:
  - Robust covariance estimation (e.g., Minimum Covariance Determinant).
  - Diagonal loading / shrinkage: $\Sigma \leftarrow \Sigma + \lambda I$ with $\lambda > 0$.
  - Feature reduction (dropping or combining highly collinear features).

11.4 Rationale

The manifold defined by (μ, Σ) represents the empirical support of legitimate residual behavior. Under IRP, impostors are expected to occupy a constrained subset of behaviors and therefore produce residual vectors R that fall outside this support.

12 Rigidity Scoring and Decision Rules

This section defines the statistical scoring procedure used to determine whether a trajectory segment is consistent with the baseline legitimacy manifold and consistent with rigidity direction under IRP.

12.1 Mahalanobis Distance Score

Distributional Assumption. Mahalanobis distance implicitly assumes an approximately elliptical baseline residual distribution. RIFT does not require strict Gaussianity; however, heavy-tailed or multimodal residual distributions may reduce threshold calibration accuracy.

Let i denote a trajectory segment under evaluation.

Let $R_i \in R^d$ denote the residual feature vector for segment i (Section 10).

Let $\mu \in R^d$ and $\Sigma \in R^{d \times d}$ denote the baseline mean and covariance (Section 11.2).

Define the Mahalanobis distance:

$$D_i = (R_i - \mu)^T \Sigma^{-1} (R_i - \mu)$$

where:

- Σ^{-1} denotes the matrix inverse of Σ (or a numerically stable pseudo-inverse if required).

Interpretation. D_i measures how far R_i lies from the baseline manifold under the baseline covariance geometry. Large values indicate outliers.

12.2 Baseline-Derived Distance Threshold

Let $\{D_i^{\text{base}}\}_{i=1}^n$ be the Mahalanobis distances computed for baseline residuals.

Define the distance threshold:

$$\tau = Q_{0.99}\left(\{D_i^{\text{base}}\}\right)$$

where:

- $Q_{0.99}(\cdot)$ denotes the 99th percentile.

A segment is an **outlier** if:

$$D_i > \tau$$

12.3 Directional Rigidity Constraint

Mahalanobis distance alone detects outliers in any direction. RIFT additionally requires that the outlier direction is consistent with rigidity under IRP.

Let $w \in R^d$ be a fixed direction vector.

In RIFT v1.0, the default choice is:

$$w = [-1, -1, -1, -1, -1, -1]$$

corresponding to the expectation that rigidity is associated with reduced:

- Identity entropy (H_S),
- Entropy rate (H_{rate}),
- LZ complexity (H_{LZ}),
- Transition graph density (T_{sparse}),
- Change rate (r),
- RF coupling (I_{coupling}).

Define the directional score:

$$\rho_i = w^T R_i$$

Interpretation. More negative ρ_i indicates a residual vector aligned with the rigidity direction.

12.4 Baseline-Derived Direction Threshold

Let $\{\rho_i^{\text{base}}\}_{i=1}^n$ be baseline directional scores.

Define the direction threshold:

$$\gamma = Q_{0.05}\left(\{\rho_i^{\text{base}}\}\right)$$

where:

- $Q_{0.05}(\cdot)$ denotes the 5th percentile.

12.5 Decision Rule

A segment SHALL be flagged as **rigidity-consistent** if and only if:

$$D_i > \tau \quad \text{and} \quad \rho_i < \gamma$$

This two-part rule reduces false positives by ensuring outliers are consistent with rigidity, not merely unusual.

13 Operational Modes

RIFT supports offline and live evaluation. Both modes MUST use identical feature extraction, conditioning, and scoring procedures. The only difference is how trajectory segments are defined.

13.1 Offline Mode

In offline mode, each input artifact (e.g., a PCAP or QMDL capture) is converted into one or more trajectory segments.

Requirements:

- Implementations MUST document how a capture is mapped to trajectory segment start and end times.
- If a capture is long, implementations MAY partition it into multiple non-overlapping segments of fixed duration.

13.2 Live Sliding-Window Mode

In live mode, segments are defined by a sliding window.

Let:

- W be the window length in seconds.
- S be the step size in seconds.

At evaluation time t, define the segment:

$$\mathcal{T}_{t-W:t} = \{Y(u)\}_{u=t-W}^{t}$$

Recommended:

- $W \geq 600$ seconds (10 minutes).
- $S = 30$ seconds.

Windows that fail missingness requirements (Section 6.3) MUST be discarded or labeled invalid.

13.3 Output Requirements

Each evaluated segment MUST produce:

- Conditioning vector Z_i.
- Feature vector F_i.
- Residual vector R_i.
- Mahalanobis distance D_i.

- Directional score ρ_i.
- Decision outcome (rigidity-consistent flag).

This output is sufficient to support post-hoc analysis, visualization, and validation.

14 Reference Pseudocode

This section provides reference pseudocode that directly corresponds to the mathematical procedure defined in this specification. It is intended to guide implementers and to reduce ambiguity.

14.1 Baseline Training

Baseline training fits:

- Conditioning models $m_j(\cdot)$ for each feature dimension.
- Baseline manifold parameters (μ, Σ) in residual space.
- Decision thresholds (τ, γ) from baseline score distributions.

```
# Inputs:
#   baseline_segments: trusted segments for model fitting
# Parameters:
#   delta_t = 1.0            # time alignment interval (seconds)
#   K = 8                    # MI discretization levels for SINR (quantiles)
#   alpha = 0.99             # distance threshold percentile
#   beta = 0.05              # directional threshold percentile
#   w = [-1, -1, -1, -1, -1, -1]

# 1) Extract (F, Z) pairs from baseline segments
rows = []
for seg in baseline_segments:
    seg = align_time(seg, delta_t=delta_t)          # Time Alignment
    S = extract_identity_sequence(seg)              # Identity Symbol Constructio
    Z = compute_conditioning(seg)                   # RF Metrics and Conditioning
    F = compute_features(S, seg.rf, K=K)            # Feature Extraction
    rows.append((F, Z))

# 2) Fit conditioning models m_j: Z -> F_j using baseline data only
models = fit_conditioning_models(rows)              # Continuous Conditioning
```

```
# 3) Compute baseline residuals R = F - m(Z)
R_list = []
for (F, Z) in rows:
    R = residualize(F, Z, models)
    R_list.append(R)

mu = mean_vector(R_list)                          # Legitimacy Manifold Modelin
Sigma = covariance_matrix(R_list)                 # Legitimacy Manifold Modelin
Sigma = stabilize_or_robustify(Sigma)             # Legitimacy Manifold Modelin

# 4) Compute baseline scores and thresholds
D_base = [mahalanobis(R, mu, Sigma) for R in R_list]  # Rigidity Scoring and Deci
tau = percentile(D_base, alpha)

rho_base = [dot(w, R) for R in R_list]            # Rigidity Scoring and Decisi
gamma = percentile(rho_base, beta)

# Persist:
#   models, mu, Sigma, tau, gamma, w
```

14.2 Evaluation (Offline or Sliding Window)

Evaluation produces a score per segment using the fixed baseline parameters learned above.

```
# Inputs:
#   segment: a trajectory segment (offline file or live window)
#   models, mu, Sigma, tau, gamma, w: trained baseline parameters

segment = align_time(segment, delta_t=delta_t)    # Time Alignment
S = extract_identity_sequence(segment)            # Identity Symbol Constructio
Z = compute_conditioning(segment)                 # RF Metrics and Conditioning
F = compute_features(S, segment.rf, K=K)          # Feature Extraction

R = residualize(F, Z, models)                     # Continuous Conditioning
D = mahalanobis(R, mu, Sigma)                     # Rigidity Scoring and Decisi
rho = dot(w, R)                                   # Rigidity Scoring and Decisi

flag_rigid = (D > tau) and (rho < gamma)          # Decision rule
```

14.3 Implementation Notes

- **Mutual information estimation:** The pseudocode assumes ΔSINR is discretized into K quantile bins per segment before empirical mutual

information is computed with ΔS.

- **Covariance stability:** Implementations MUST handle singular or ill-conditioned Σ as described in Section 11.3.
- **Determinism:** Model fitting and quantile binning SHOULD be deterministic given the same baseline dataset to support reproducibility.

15 Security Considerations

RIFT is a detection protocol based on empirical modeling of legitimate trajectory evidence. Correctness depends on baseline integrity and on the assumption that an adversary cannot perfectly reproduce the full range of legitimate trajectory variability under exogenous coupling.

15.1 Baseline Integrity

Baseline segments MUST be collected from environments where the analyst has a high degree of confidence that:

- The serving network infrastructure is legitimate.
- The receiver is not being actively targeted by a sophisticated impostor during baseline collection.

If baseline segments are contaminated, the legitimacy manifold will be corrupted and detection performance will degrade.

15.2 Adversary Adaptation

RIFT is designed to reduce reliance on brittle heuristics. However, a sufficiently capable adversary may attempt to approximate legitimate trajectory structure.

Implementations SHOULD treat RIFT scores as evidence to be corroborated by additional signals, including:

- Traditional rogue-cell heuristics.
- Cross-device agreement when multiple receivers are available.
- Cross-location consistency when repeated measurements are possible.

15.3 Operational Risk

RIFT output may be used in safety or security decision-making contexts. Implementations SHOULD:

- Persist all intermediate outputs (Z_i, F_i, R_i, D_i, ρ_i) for auditability.
- Avoid destructive automated actions solely based on a single flagged window.

16 Core Claim

RIFT is grounded in two supporting ideas:

- **Stateful Temporal Entropy (STE):** Legitimacy claims MUST be supported by trajectory evidence derived from receiver observations over irreversible progression, rather than by static snapshots.
- **Imposter Rigidity Principle (IRP):** Impersonation under incomplete information constrains admissible variability. An impostor is expected to exhibit structural rigidity relative to the manifold of legitimate behavior.

RIFT detects rigidity by identifying residual trajectory-structure features that deviate from the baseline legitimacy manifold after conditioning on RF regime.

17 References

- Shannon, C. E. “A Mathematical Theory of Communication.” 1948.
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- Curnow, M. “Stateful Temporal Entropy: A Synthesis of Trajectory, Irreversibility, and Evidence.” 2025.
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18 Appendices

18.1 Appendix A - Changelog

Date	Change/Comments	Version
3/1/2026	Version 1.0 drafted.	Version 1.0

18.2 Appendix B - Threat Model

This appendix describes the assumed adversary model, detection assumptions, and scope limitations of RIFT. The purpose of this section is to clarify the security boundaries under which RIFT is intended to operate.

18.2.1 Adversary Capabilities

RIFT assumes an adversary capable of:

- Deploying rogue cellular infrastructure.
- Broadcasting valid-looking identity parameters (e.g., PLMN, TAC, PCI, Cell ID).
- Maintaining stable RF transmission over time.
- Reconfiguring broadcast parameters to avoid static heuristic detection.

RIFT does not assume the adversary can:

- Perfectly reproduce the full statistical variability of legitimate trajectory behavior under diverse environmental coupling.
- Simultaneously emulate multi-site spatial diversity across independent receivers.
- Access, corrupt, or poison the trusted baseline dataset used for manifold construction.

18.2.2 Detection Assumption

RIFT operates under the following assumptions:

1. Legitimate broadcast identity trajectories exhibit structured variability across time.

2. Impersonation under incomplete information constrains admissible variability.
3. These constraints produce detectable residual rigidity after RF conditioning.

18.2.3 Out-of-Scope Adversaries

RIFT does not claim to detect:

- Perfect statistical emulators trained on identical baseline data capable of reproducing the residual feature distribution.
- Adversaries with complete knowledge of conditioning models and legitimacy manifold geometry who can reproduce residual distributions across all feature dimensions.
- Compromised receivers producing falsified or manipulated observation streams.

18.2.4 Multi-Receiver Strengthening (Optional Extension)

Detection confidence increases when:

- Multiple independent receivers observe correlated rigidity patterns.
- Spatially separated trajectory segments exhibit synchronized structural constraints.

Multi-receiver corroboration reduces the likelihood that detected rigidity is attributable to localized RF artifacts or receiver-specific behavior.